Testing notes

- interfaces - enp7s0
- MAC attacker - 16:53:d2:bc:4c:91
- MAC victim - 1a:5e:73:5b:e3:9b
- victim userspace:

```
sudo python3 scripts/memwatch.py --out results/userspace-test-100x -- ./detector-userspace -iface enp7s0 -window 536.870912ms -duration 60m --out results/userspace-test-100x -block
```

- victim xdp with blocking:

```
sudo python3 scripts/memwatch.py --out results/xdp-test-15x --bpftool -- ./detector-xdp -iface enp7s0 -window 536.870912ms -duration 60m -out results/xdp-test-15x -block
```

- crazy jump in userspace:
  - detector: 6m14s: 190870 packets, 5429 dropped by the socket, 739 windows (18 flagged)
  - detector: 6m24s: 291609 packets, 23058 dropped by the socket, 759 windows (20 flagged)
- remember:
  - to block or not to block
  - window size

```
Dec  1, 2018 13:17:10.711517000 UTC
Dec  1, 2018 14:22:10.711517000 UTC
```

```
python3 scripts/excerpt.py --input /mnt/attack_data/DAT300-data/CICDDoS2019/PCAP/01-12/PCAP-01-12_0-249/ --out /mnt/attack_data/DAT300-data/excerpts/first-excerpt.pcap --duration 105m --start 2018-12-01T14:22:10Z
```

```
scp fabvm:/mnt/.... .
```

/mnt/attack_data/DAT300-data/CICDDoS2019/PCAP/01-12/PCAP-01-12_0-249

- attack starts at the end of 0
- 195 it slowes down
- last quarter of 0 till 195 (It's from 2018-12-01 14:21:52.153161000 to 2018-12-01 15:03:06.989875)
- scenarios (if we have time do each 3 times):
  - real-time (about 40 minutes each)
    - accuracy analysis - unblocked:

- userspace blocklist xdp generic

```
# 1. Compile
clang -O2 -g -target bpf -I/usr/include/$(uname -m)-linux-gnu -c blocklist.c -o
blocklist.o
# 2. Attach
sudo ip link set dev enp7s0 xdpgeneric obj blocklist.o sec xdp
# 3. Double check the attach
ip link show
# 4. Run the test
sudo python3 scripts/memwatch.py --out results/usr-acc-gen-1 -- ./detector-userspace -
iface enp7s0 -window 536.870912ms -duration 60m --out results/usr-acc-gen-1
# 5. Detach
sudo ip link set dev enp7s0 xdpgeneric off
# 6. Delete all the pinned bpf maps
for dir in /sys/fs/bpf /sys/fs/bpf/tc/globals; do
    sudo rm -f --
"$dir"/{blocklist_map,sketch_{0..3},topk_map_{0..3},slot_stats,blocked_stats,us_blockli
st_map,us_blocked_stats}
done
```

- userspace blocklist xdp native (FABRIC)

```
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clang -O2 -g -target bpf -I/usr/include/$(uname -m)-linux-gnu -c blocklist.c -o
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# 2. Attach
sudo ip link set dev enp7s0 xdpnative obj blocklist.o sec xdp
# 3. Double check the attach
ip link show
# 4. Run the test
sudo python3 scripts/memwatch.py --out results/usr-acc-nat-1 -- ./detector-userspace -
iface enp7s0 -window 536.870912ms -duration 60m --out results/usr-acc-nat-1
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st_map,us_blocked_stats}
done
```

- kernel generic

```
# 1. Compile
clang -O2 -g -target bpf -I/usr/include/$(uname -m)-linux-gnu -c heavy_keeper.c -o
heavy_keeper.o
```

```
# 2. Attach
sudo ip link set dev enp7s0 xdpgeneric obj heavy_keeper.o sec xdp
# 3. Double check the attach
ip link show
# 4. Run the test
sudo python3 scripts/memwatch.py --out results/ker-acc-gen-1 --bpftool -- ./detector-
xdp -iface enp7s0 -window 536.870912ms -duration 60m -out results/ker-acc-gen-1
# 5. Detach
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```

- performace metrics - blocked (not only -block flag but also --cpu):
  - userspace blocklist xdp generic

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# 3. Double check the attach
ip link show
# 4. Run the test
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```
sudo python3 scripts/memwatch.py --cpu --out results/usr-per-gen-1 -- ./detector-
userspace -iface enp7s0 -window 536.870912ms -duration 60m --out results/usr-per-gen-1
-block
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# 4. Run the test
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heavy_keeper.o
# 2. Attach
sudo ip link set dev enp7s0 xdpgeneric obj heavy_keeper.o sec xdp
# 3. Double check the attach
ip link show
# 4. Run the test
sudo python3 scripts/memwatch.py --cpu --out results/ker-acc-per-1 --bpftool --
./detector-xdp -iface enp7s0 -window 536.870912ms -duration 60m -out results/ker-acc-
per-1 -block
# 5. Detach
```

```
sudo ip link set dev enp7s0 xdpgeneric off
# 6. Delete all the pinned bpf maps
for dir in /sys/fs/bpf /sys/fs/bpf/tc/globals; do
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st_map,us_blocked_stats}
done
```

- maybe sped up (maybe whole dataset)
- attack:

```
#!/bin/bash
MTU=1500

# subject to change:
ATTACKER_DEV=enp7s0
VICTIM_MAC="1a:5e:73:5b:e3:9b"
DATA="/mnt/attack_data/DAT300-data/CICDDoS2019/PCAP/01-12/PCAP-01-12_0-249/" # absolute
path

find "$DATA" -maxdepth 1 -type f -print0 |
  sort -zV > ~/replay-results/files.nul

mapfile -d '' -t PCAPS < ~/replay-results/files.nul

printf 'Number of files: %s\n' "${#PCAPS[@]}"
```

```
printf '%s\n' "${PCAPS[@]:0:5}"

REPLAY_OPTIONS=(
  --intf1="$ATTACKER_DEV"
  --dlt=enet
  --enet-smac="$ATTACKER_MAC"
  --enet-dmac="$VICTIM_MAC"
  --mtu="$MTU"
  --mtu-trunc
  --fixcsum
  --no-flow-stats
)

for pcap in "${PCAPS[@]}"; do
  printf '\nReplaying: %s\n' "$pcap"

  sudo tcpreplay-edit \
    "${REPLAY_OPTIONS[@]}" \
    "$pcap" || break
done
```